Land Cover & Soil Moisture Analysis

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Table of Contents

Abstract 3

2. Introduction 4

3. Soil moisture methodology 6

3.1 STUDY AREA 6
3.2 DATA SOURCES 7
3.3 DATA LIMITATIONS 8
3.4 SPECIFIC PRE-PROCESSING TECHNIQUES 8
3.5 EQUATIONS 10

4. Soil moisture results 11

4.1 SOIL MOISTURE ANALYSIS 11
4.2 PRECIPITATION TRENDS 15
4.3 SEASONAL DIFFERENCE ANALYSIS 16
4.3.1 across satellite missions 17
4.3.2 across regions 18
4.3.3 statistical analysis 19

5. Discussion 20

5.1 RESULTS INTERPRETATION 20
5.2 PREVIOUS WORK 21
5.3 ACCURACY ERRORS 21

6. Conclusion 21

References 23

Abstract

This study investigates soil moisture dynamics and land use/land cover (LULC) change in Sudan with emphasis on the Red Sea, Khartoum, and South Kordofan regions. Using satellite-derived datasets from MODIS, and SMAP, combined with indices such as the Standardized Soil Moisture Index (SSI), the analysis covers the period 2016-2024. The results highlight significant interannual and seasonal variability in soil moisture, with dry anomalies observed in 2017 and 2024 and wet anomalies in 2018 and 2021. Comparisons between MODIS and SMAP reveal consistent spatial and temporal patterns, although differences in anomaly magnitudes are evident. Statistical assessments, including bias, R^2, and correlation, confirm the robustness of the datasets and their suitability for monitoring soil moisture in semi-arid regions. The findings underscore the role of remote sensing in quantifying land degradation, monitoring desertification, and evaluating the effectiveness of climate adaptation strategies such as the Great Green Wall initiative.

Keywords: Soil moisture, Remote sensing, MODIS, SMAP, Sudan, SSI, Desertification

2. Introduction

Land is a critical resource that underpins many human activities, including agriculture, manufacturing, forestry, energy generation, housing, leisure and water management. Land has historically driven economic growth due to its natural characteristics, which include soil quality, topography, climate, geology, hydrology, biodiversity and political divisions, in addition to socioeconomic factors such as technological advancements and effective land management practices (Abualgasim et al., 2022).

The Sudano-Sahelian region, a semi-arid area in Central Africa between latitudes 13° 7' and 17° 59' N (Figure 1), is becoming more prone to desertification and extreme weather events, such as droughts. These conditions have had a significant influence on local economies and crucial development sectors (Alriah et al., 2024). As a result, land restoration programs such as the African Union's Great Green Wall (GGW) initiative begun in 2007, which has become critical for improving climate resilience, combatting desertification, and promoting sustainable development (Deng et al., 2024).

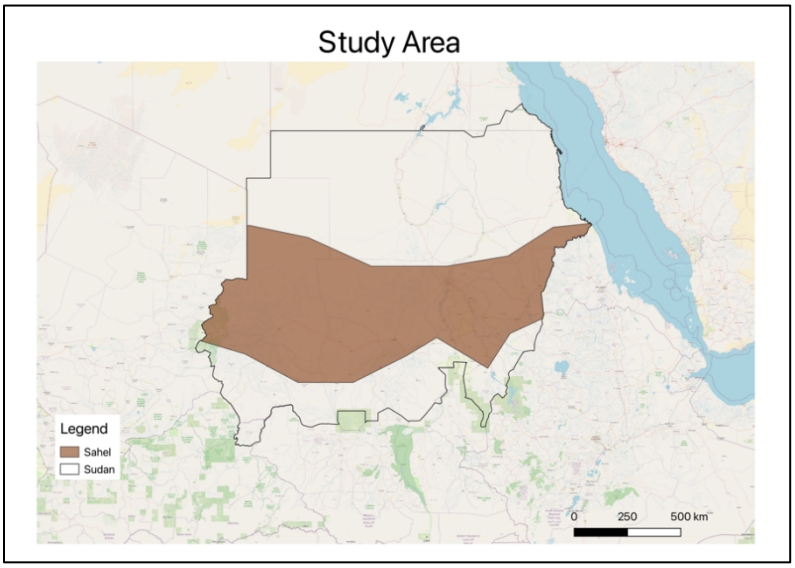


Figure 1

Sudan-Sahel Region

The region experiences the most rainfall between July and August during the wet season which spans from June to September. The dry season runs from November to May.

Significant gaps have emerged in the literature as it has advanced. With the research focused on Sudan-Sahel, these studies have mostly focused on Sub-Saharan West Africa and the Horn of Africa to the east (Yang et al., 2022). Furthermore, much of literature has concentrated on land degradation caused by erosion, whereas more study is needed to address other significant issues such as soil moisture depletion and long-term time-series vegetation transitions.

Understanding LULC variations can help research integrate environmental factors that influence soil moisture, vegetation, surface water, and urban land (Huber et al., 2011).

Remote sensing, especially satellite-based change detection, has shown great potential for effectively, monitoring land cover changes. This involves comparing images of the Earth's surface taken at different times to detect and understand environmental transformations over extensive areas (Singh, 1989; Abualgasim et al., 2022). Despite its advantages, there's still limited research specifically focusing on monitoring progress and impacts in the GGW region, highlighting an important gap in our understanding of these areas (Deng et al., 2024).

Furthermore, improving knowledge about vegetation recovery patterns in this region is critical, as it could offer valuable insights into how people manage their movements and recourses. It would also enhance the ability to model complex ecological systems (Kumar et al., 2002). Therefore, this study aims to demonstrate how remote sensing techniques can be used effectively to investigate soil moisture changes, desertification patterns, fluctuations in surface water, and urban displacement over the past 20 to 8 years. LULC is assessed for north, central and south regions of Sudan to demonstrate patterns mostly focusing on Khartoum in the central as it is in the transition zone. The study by Yang et al., (2022) highlights the technology developments in remote sensing applications and their consequences for sustainable management plans and environmental monitoring. Strengthening land degradation mitigation techniques in the Sudan-Sahel region would need to encompass current research gaps and utilize innovative approaches.

The main objectives of this research area to:

1. Assess the relation between vegetation cover, soil moisture, surface water, and urban development.
2. Integrate satellite datasets from different spatial and temporal resolutions.
3. Identify key factors driving LULC changes for north, central and south regions of Sudan

Satellite missions such as Landsat, Moderate-resolution Imaging Spectroradiometer (MODIS), and Soil Moisture Active Passive (SMAP), provide valuable data for tracking environmental trends. To combine these datasets effectively, careful data preprocessing and calibration are essential. Remote sensing indices such as the Standardized Soil Moisture Index (SSI), will be applied to analyse these changes comprehensively.

Addressing LULC changes is critical for solving global environmental concerns, given the extensive human-caused changes that contribute to deforestation, biodiversity loss, global warming and desertification. Integrating remote sensing data with information on population growth and demographic trends is critical for making informed choices and managing resources sustainably (Abualgasim et al., 2022). Furthermore, the fast growth of human activity poses continual challenges to environmental and biological systems. A decades-worth of remote

sensing studies provide accurate and precise insights into these large-scale environmental changes, outperforming short-term, localized investigations.

3. Soil Moisture Methodology

3.1 Study Area

Soil moisture assessment is performed for three different regions within Sudan to properly assess the patterns throughout the Country. Chosen regions are Red Sea (north), Khartoum (Central), and South Kordofan (South). See Figure 2. The reason behind this is to examine the shifts of dry and wet seasonal changes, distinct ecological zones and relation to desertification.

Red Sea covers an area of 212,800 square kilometres and has population of 1,566,808 inhabitants. It mostly experiences desert climate extreme hot summer and relatively hot winters. Over 90% of rainfall here primarily occurs in November. The average annual rainfall is 76 millimetres, while average temperature is 28.4 degrees Celsius (Sudan unicef, n.d.). In addition, Khartoum is home for 6,334,348 people and cover, 1,010 square kilometres. It is subtropical desert with moderate wet season between July and September. In the Winter, from December to February, it hot during the day but cold at nights. With yearly average temperature of 30.5 degrees Celsius and precipitation of 120 millimetres, it is one of the world's hottest capitals (Khartoum climate, n.d.). Lastly, South Kordofan has population of 2,000,000 and growing every year, it covers an area of 120,000 square kilometres. The region has subtropical steppe climate with an average temperature of 30.23 degrees Celsius and precipitation about 104 millimetres (South Kordofan, 2015).

The focus behind these study areas is to understand if Red Sea regions experience extremely dry soil moisture conditions, if Khartoum experiences varying conditions depending on various factors, and if South Kordofan experiences higher rainfall. This provides deeper insights into soil moisture dynamics. Moreover, understanding of how soil moisture variations can influence environmental stability within the region.

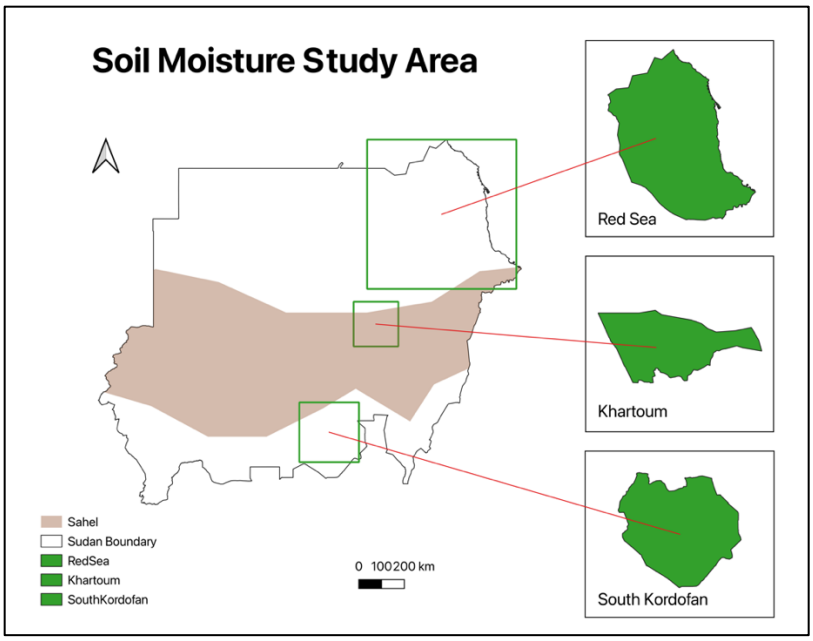


Figure 2

Three regions of soil moisture study area are Red Sea, Khartoum, South Kordofan

3.2 Data Sources

The satellite imagery used for soil moisture assessment is acquired from MODIS and SMAP sensors from 2016 to 2024. MODIS, which uses optical sensors, provides valuable data on land surface temperatures at a spatial resolution of 1000 meters and daily temporal resolution (MOD11A1 product). The bands used from this product are LST_Day_1km and QC_Day. SMAP, equipped with passive microwave sensors, provides precise soil moisture data at spatial resolutions of 11000 meters, updated every three hours (SPL4SMGP 007 product). The bands used from this product are sm_surface. This dataset specifically measures moisture content in the top 0-5 centimetres of soil, offering continuous and accurate moisture tracking. Land surface temperature is closely related to soil moisture and thus serves as reliable indicator for analysis.

Additionally, precipitation data is acquired from the Global Precipitation Measurement (GPM) mission, specifically the IMERG V07 product, which integrates data from multiple satellites by merging all passive-microwave sensors in the GPM constellations to produce consistent global precipitation observations. The dataset has a spatial resolution of 11,132 meters and updates every 30 mins. The band used is precipitation, which provides calibrated images.

Data retrieval, processing and analysis were conduct using the Python API of Google Earth Engine through Jupyter Notebook. Key Python libraries used include ee and geemap for data acquisition and visualization as well as pandas, matplotlib, sklearn.metrics and suborns for subsequent data processing, and statistical visualization. Additionally, Excel was used to conduct the accuracy assessment.

3.3 Data Limitations

One of the major constraints of using MODIS and SMAP data is their different spatial resolutions. While MODIS provides great temporal resolution, its thermal bands have very low geographical resolution (Deng et al., 2024). Moreover, accurate terrain adjustments are not feasible due to the unavailability of complete worldwide high-resolutions Digital Elevation Model (DEM) and limitation in MODIS positioning accuracy. Thus, simplified surface elevation data from MOD02 of MODIS are used in land surface temperature models.

In contrast, ground-based station observations measure soil moisture precisely at very localized scale, capturing only small volumes of soil, while satellite sensors aggregate data over large footprints, such as SMAP's near 11,000-meter resolution. This aggregation may confound direct comparisons and reduce accuracy evaluations of satellite data (Xu et al., 2018). The accuracy of soil moisture retrieval is greatly impacted by vegetation, especially when passive microwave sensors and optical sensors are used. Dense vegetation can mask soil signals, causing moisture levels to be overestimated or underestimated (Huber et al., 2011; Abualgasim et al., 2022; Javed et al., 2019).

Lastly, IMERG's data collected from numerous satellite sensors of varied availability lacks temporal consistency, promoting concerns when applied to climate-scale research, as the approach favours high-resolution data over uniform temporal values.

3.4 Specific Pre-Processing Techniques

Data from MOD11A1 underwent a cloud masking process to exclude pixels affected by cloud coverage, explicitly using QC_Day bands to select values with a QC equal to 0. Both MOD11A1 (LST) and SPL4SMGP 007 (sm surface) pixel values were normalized, with MODIS values converted to Celsius from Kelvin. Normalizing pixel values between a minimum of 0 and a maximum of 1 helped ensure consistency, particularly important when resampling data, as it involves combining multiple pixel values. The following equation (Equation 1) illustrates Celsius conversion of MODIS LST values:

$$\text{mean}\left(\text{LST}_{\text{DAY}_{\text{1KM}}\text{yearly}}\right) \times 0.02 - 273.15 \quad (1)$$

Where Lst_Day_1KM indicated MODIS band used and numerical values to convert to Celsius.

The following equations (Equation 1 & 2) show normalization of MODIS and SMAP values:

$$\frac{MODIS_{Celsius} - Min_{MODIS-Celsius}}{Max_{MODIS_{Celsius}} - Min_{MODIS_{Celsius}}} \quad (2)$$

$$\frac{sm_{surface} - Min_{sm_{surface}}}{Max_{sm_{surface}} - Min_{sm_{surface}}} \quad (3)$$

The MODIS Celsius conversion, calculated using the previous equation (Equation 1), was used to determine the minimum and maximum values of the derived Celsius data. For SMAP normalization, the *sm_surface* band was applied, with the minimum and maximum values corresponding to the lowest and highest band values, respectively.

Initially, data was tested without normalization, but due to significant differences in spatial resolution between MODIS and SMAP pixels, normalization was necessary. Prior to analysis, MODIS data were resampled to match the spatial resolution of SMAP data using bilinear interpolation. This upscaling process was advantageous because it enhanced the matching accuracy between MODIS and SMAP datasets by incorporating multiple original MODIS pixels into each SMAP-scale pixel. Bilinear interpolation was elected due to its ability to produce smoother and more accurate results compared to nearest-neighbour interpolation, as it uses surrounding pixel values to estimate resampled pixels effectively. Root Mean Square Error (RMSE) computations were used to evaluate the consistency of MODIS and SMAP data, comparing how MODIS pixel values correlated to SMAP. Similarly, precipitation data from GPM missions (IMERG V07) was normalized and downscaled through bilinear interpolation to align spatially with the SMAP dataset.

For these datasets, time-series graphs were extracted for yearly seasonal changes in three study areas. The times series analysis was conducted to examine the difference between MODIS and SMAP satellite missions for the 2016-2024 period. In addition, image differencing was conducted for change detection analysis between 2016 and 2024 for MODIS and SMAP data and to validate time series analysis results. This analysis is to demonstrate the difference between satellite mission and yearly seasonal changes. The layers were extracted and visualized on QGIS software with scale between -1 (min) and 1 (max) values. These pixels are classified as low soil moisture to high soil moisture to visualize. Refer to table below for colour values:

Table 1

Shows how pixels are visualized for image differencing

Colour	Classification
Brown	Very low
Orange	Low
Yellow	Moderate
Green	High
Blue	Very high

Further accuracy evaluation was carried out in Excel, examining seasonal variations (February – October) in MODIS, SMAP and GPM values throughout the three study regions. The obtained difference values were normalized between -1 and 1, allowing for direct comparison across datasets. Statistical analysis, such as bias, R^2 scores and correlation coefficient, were used to compared and interpret soil moisture difference between MODIS and SMAP data across area and seasons.

3.5 Equations

For MODIS and SMAP satellite data, Temperature Moisture-Dryness Index (TMDI) and Standardized Soil Moisture Index (SSI) were used to estimate soil moisture specifically for the months of October and February from 2016 to 2024. These values area clipped to 1 and -1 as the threshold. As MODIS does not directly measure soil moisture anomalies, TMDI, derived from normalized LST, serves as an effective alternative for this. Temperature-Vegetation Dryness Index (TVDI). TMDI provides an accurate representation of drought conditions over space and time (Thai & Lois, 2024). It is important to note that this normalization for TMSI is distinct from initial normalization performed before resampling, as it specifically highlights variations in dryness and moisture conditions.

Using TMDI derived from MODIS data, the SSI was calculated to further assess soil moisture conditions from both MODIS and SMAP datasets. The SSI computation involved extracting mean and standard deviation values to standardize soil moisture content. The following equations (Equations 4, 5, and 6) illustrate the remote sensing index calculations:

$$\frac{\mathrm{LST_{TMDI}} - \mathrm{Min_{LST_{TMDI}}}}{\mathrm{Max_{LST_{TMDI}}} - \mathrm{Min_{LST_{TMDI}}}} \quad (4)$$

$$\frac{LST_{TMDI} - Mean_{LST_{TMDI}}}{Std_{LST_{TMDI}}} \quad (5)$$

$$\frac{sm_{surface} - Mean_{sm_{surface}}}{std_{sm_{surface}}} \quad (6)$$

Where *LST_TMDI* represents the band value as well as the minimum and maximum values of the band. For MODIS SSI and SMAP SSI, LST_TMDI corresponds to the index calculated from TMDI using the mean and standard deviation of the values. Similarly, for *sm_surface*, the band represents surface soil moisture and its mean and standard deviation values.

Additionally, the Standardized Precipitation Index (SPI) was calculated using precipitation data (see Figure 6) from GPM missions for the same periods as MODIS and SMAP. Including precipitation data helps determine reliability and accuracy of satellite-derived soil moisture measurements during different seasonal conditions. The following equations (Equations 7 and 8) is used normalized and calculate GPM precipitation values:

$$\frac{\text{percipitation}_{\text{norm}} - \text{Min}_{\text{percipitation}_{\text{norm}}}}{\text{Max}_{\text{percipitation}_{\text{norm}}} - \text{Min}_{\text{percipitation}_{\text{norm}}}} \quad (7)$$

$$\frac{\text{percipitation}_{\text{norm}} - \text{Mean}_{\text{percipitation}_{\text{norm}}}}{\text{std}_{\text{percipitation}_{\text{norm}}}} \quad (8)$$

Where precipitation is the band and minimum and maximum values of the band. The precipitation_norm is the normalized band values and mean and standard deviation values of the band.

The following equation (Equation 9) was used to normalize seasonal differences of SSI values and assess accuracy using Excel:

$$z = 2 \times \frac{\text{SSI}_{\text{Feb}} - \text{SSI}_{\text{Oct}} - \min\left(\text{SSI}_{\text{Feb}} - \text{SSI}_{\text{Oct}}\right)}{\max\left(\text{SSI}_{\text{Feb}} - \text{SSI}_{\text{Oct}}\right) - \min\left(\text{SSI}_{\text{Feb}} - \text{SSI}_{\text{Oct}}\right)} - 1 \quad (9)$$

Where SSI_Feb was the calculated SSI values for February and SSI_Oct was the calculated SSI value for October. The minimum and maximum values were multiplied by 2 and subtracted by 1.

The following equations (Equations 10, 11 and 12) show statistical calculation of bias, R^2, and correlation:

$$\text{Avg}\left(\text{Range}\left(\text{MODIS}_{\text{SSI}_{\text{Seasonal}_{\text{norm}_{\text{Diff}}}}} - \text{SMAP}_{\text{SSI}_{\text{Seasonal}_{\text{Norm}_{\text{Diff}}}}}\right)\right) \quad (10)$$

$$RSQ\left(\text{Range}\left(\text{MODIS}_{\text{SSI}_{\text{Seasonal}_{\text{Norm}_{\text{Diff}}}}}\right), \left(\text{SMAP}_{\text{SSI}_{\text{Seasonal}_{\text{Norm}_{\text{Diff}}}}}\right)\right) \quad (11)$$

$$\text{CORREL}\left(\text{Range}\left(\text{MODIS}_{\text{SSI}_{\text{Seasonal}_{\text{Norm}_{\text{Diff}}}}}\right), \left(\text{SMAP}_{\text{SSI}_{\text{Seasonal}_{\text{Norm}_{\text{Diff}}}}}\right)\right) \quad (12)$$

Where MODIS_SSI_Seasonal_Norm_Diff is the calculated normalized difference of MODIS values between February and October and SMAP_SSI_Seasonal_Normal_Diff is the calculated normalized difference of SMAP values between February and October.

4. Soil Moisture Results

4.1 Soil Moisture Analysis

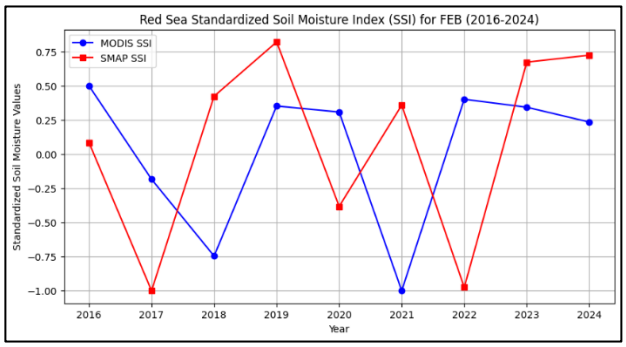
Red Sea soil moisture in February (Figure 4a and Figure 4b) shows significantly low values, indicating dry season, the graph shows high variability, likely due to the SSI index's sensitivity to dry conditions, particularly in SMAP. MODIS shows peak drops in 2018 and 2022, while SMAP shows them in 2017 and 2021. For 2023 and 2024, MODIS and SMAP show an increase in values, indicating positive trends for the regions. October Red Sea time series graph shows significant decrease in values for period. Looking at specific years, MODIS indicates a

peak high in 2017 and SMAP in 2021, while the peak low for MODIS is in 2024 and for SMAP in 2023.

For Khartoum (Figure 4c and Figure 4d), the dry season during February shows an overall downward trend from 2016 to 2024 in contrast to the Red Sea in February. SMAP shows a smoother decline, while MODIS displays less variability than in the Red Sea. The peak was in 2016 for MODIS and 2018 for SMAP. During 2021, the values are approximately the same for MODIS and SMAP, indicating subtle commonalities between the satellites. October values remain relatively stable, with highs and lows changing gradually over time. The values for 2019 and 2023 are relatively similar. The peak was during 2017 for MODIS and 2019 for SMAP, while low was in 2024 for MODIS and 2017 for SMAP.

Lastly, South Kordofan (Figure 4e and Figure 4f) demonstrates steep downward pattern during February for MODIS and SMAP and values relatively are approximate from 2021 to 2023. There is an increase in values for 2024. The peak for MODIS, is in 2021 and SMAP in 2019, where it's low in 2023 for both. During October, the graph shows high variability throughout the period. There is a decrease shown in 2021 for both MODIS and SMAP. The peak was in 2020 for MODIS and 2022 for SMAP, whereas low was in 2019 for MODIS and 2021 for SMAP.

(a)



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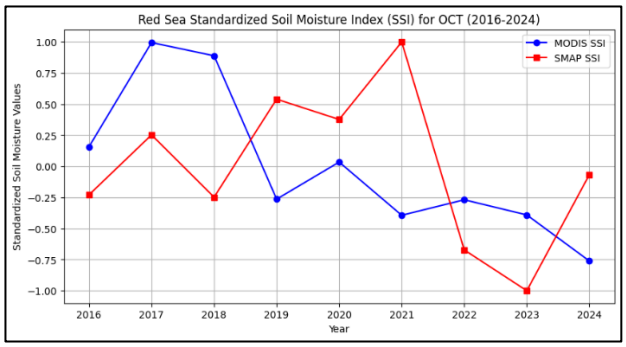

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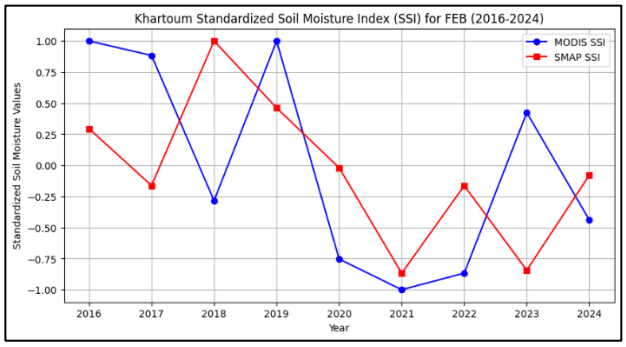

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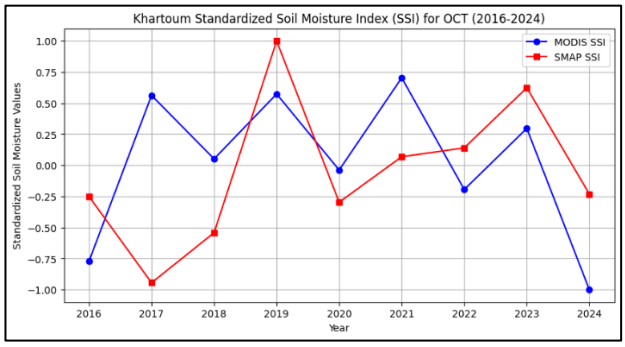

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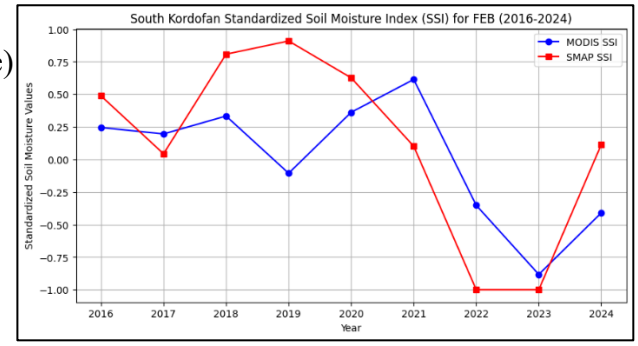

(f)

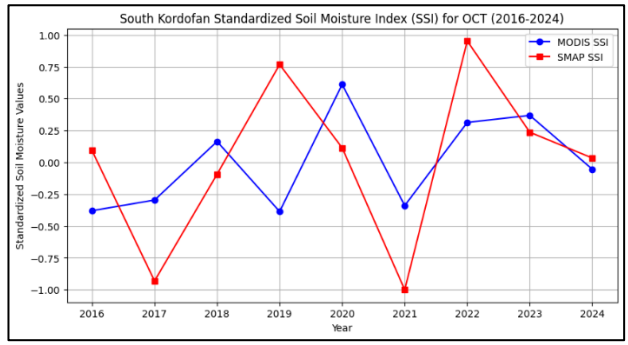

Figure 4

Time series graphs from 2016-2024 yearly SSI values for MODIS and SMAP organized regions for the months of February and October. (a) This graph presents the Standardized Soil Moisture Index (SSI) for the Red Sea region in February, covering the years 2016–2024. (b) The October SSI values for the Red Sea show a different seasonal dynamic. (c) This plot shows February SSI values for Khartoum. (d) The October SSI in Khartoum exhibits more frequent oscillations between positive and negative values compared to February. (e) February SSI for South Kordofan indicates substantial interannual variability. (f) October SSI values for South Kordofan also show alternating wet and dry years.

Figure 5 shows images differencing between 2016 and 2024 values to assess clearer patterns between two specific years.

For the month of February, MODIS data shows a decrease in soil Moisture throughout large position of the region, implying ongoing surface drying in Red Sea (Figure 5a and Figure 5b). In contrast, SMAP exhibits a more constant and broader rise in soil moisture between 2016 and 2024, indicating its sensitivity to topsoil moisture. Additionally, for wet season during October there is a significant drying for the image difference 2016-2024 mostly showing brown pixels.

In Khartoum, both MODIS and SMAP images for February show an overall decline in soil moisture from 2016-2024 images (Figure 5c and Figure 5d). The individual images for 2016 and 2024 correspond well the time series values, validating the trend. Contrary to February, October shows mixed patterns. Both increases and decreases are observed in MODIS and SMAP image difference. For SMAP, the 2016-2024 image shows more blue pixels. For MODIS, there is a decrease in brown pixels in 2024 compared to 2016-2024, through there is no significant increase in blue pixels.

Similarly, in South Kordofan, both MODIS and SMAP show an overall decrease in soil moisture in the February 2016-2014 image difference (Figure 5e and Figure 5f). Even though SMAP data indicates a slight increase in soil moisture in 2024, the overall image difference still shows more brown pixels. In October, MODIS shows an increase in soil moisture in 2016-2024 image, with more blue pixels than brown in 2024, although brown pixels remain relatively

unchanged. SMAP shows both increase and decrease in October, with a mix of blue and brown pixels across the region. This suggests localized moisture gain in some area and drying in others.

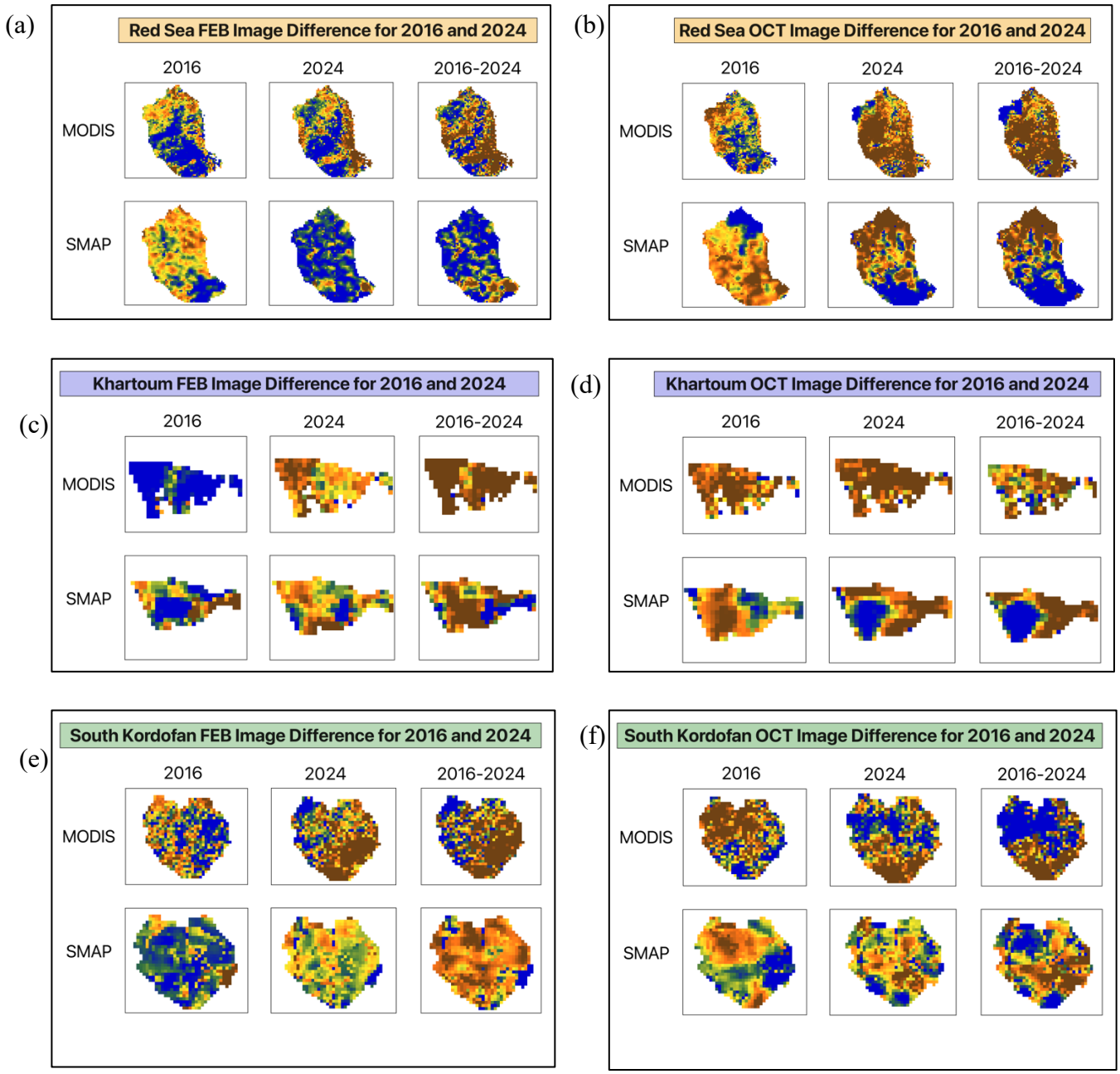

Figure 5

Image differencing between 2016 and 2024 for yearly SSI values organized by regions for February and October months. (a) and (b) Red Sea February and October SSI anomaly image differences respectively from MODIS and SMAP for 2016, 2024, and the long-term change (2016–2024). (c) and (d) Khartoum February and October SSI anomaly image differences respectively from MODIS and SMAP for 2016, 2024, and the long-term change (2016–2024). (e)

and (f) South Kordofan February and October SSI anomaly image differences respectively from MODIS and SMAP for 2016, 2024, and the long-term change (2016–2024).

Comparing the results of time series analysis and image difference spatial patterns to RMSE (Table 2). the values show how accurate MODIS data is to SMAP. South Kordofan shows overall low error. While Khartoum October show extremely high error indicating no pattern between MODIS and SMAP and is supported by image difference figure of this area.

Table 2

Root Mean Square Error for February and October Months calculated to statistically validate the difference in MODIS and SMAP during these months.

RMSE (%)	Month	Red Sea	Khartoum	South Kordofan
	FEB	21.78	33.96	28.2
	OCT	35.61	60.75	33.57

4.2 Precipitation Trends

The peak SPI for the Red Sea regions (Figure 7a and Figure 7b) in February occurred in 20217, but the corresponding soil moisture peaks appear in 2022 for MODIS and SMAP in 2019 according to the time series graphs. For October, the SPI peak was in 2018, while MODIS and SMAP show peak soil moisture values in 2017 and 2021 respectively. This suggests a mismatch between precipitation and soil moisture values.

In Khartoum, peak values occurred in 2019 and 202 for February (Figure 7c and Figure 7d). MODIS aligns with 2019 as the soil moisture peaks, while SMAP shows a peak in 2018. For October, both MODIS and SMAP record 2019 as the peak, matching the SPI peak. This indicates a positive relationship between precipitation and soil moisture.

For South Kordofan, the peak SPI for February was in 2022 (Figure 7e and Figure 7f), but MODIS and SMAP time series graphs show soil moisture values in that year. Indicating there is a discrepancy. In October, SPI peaked in 2019. MODIS shows 2022 as the peak soil moisture year, while SMAP identifies 2019 as one of its peak values, implying a level of agreement but also regional variability.

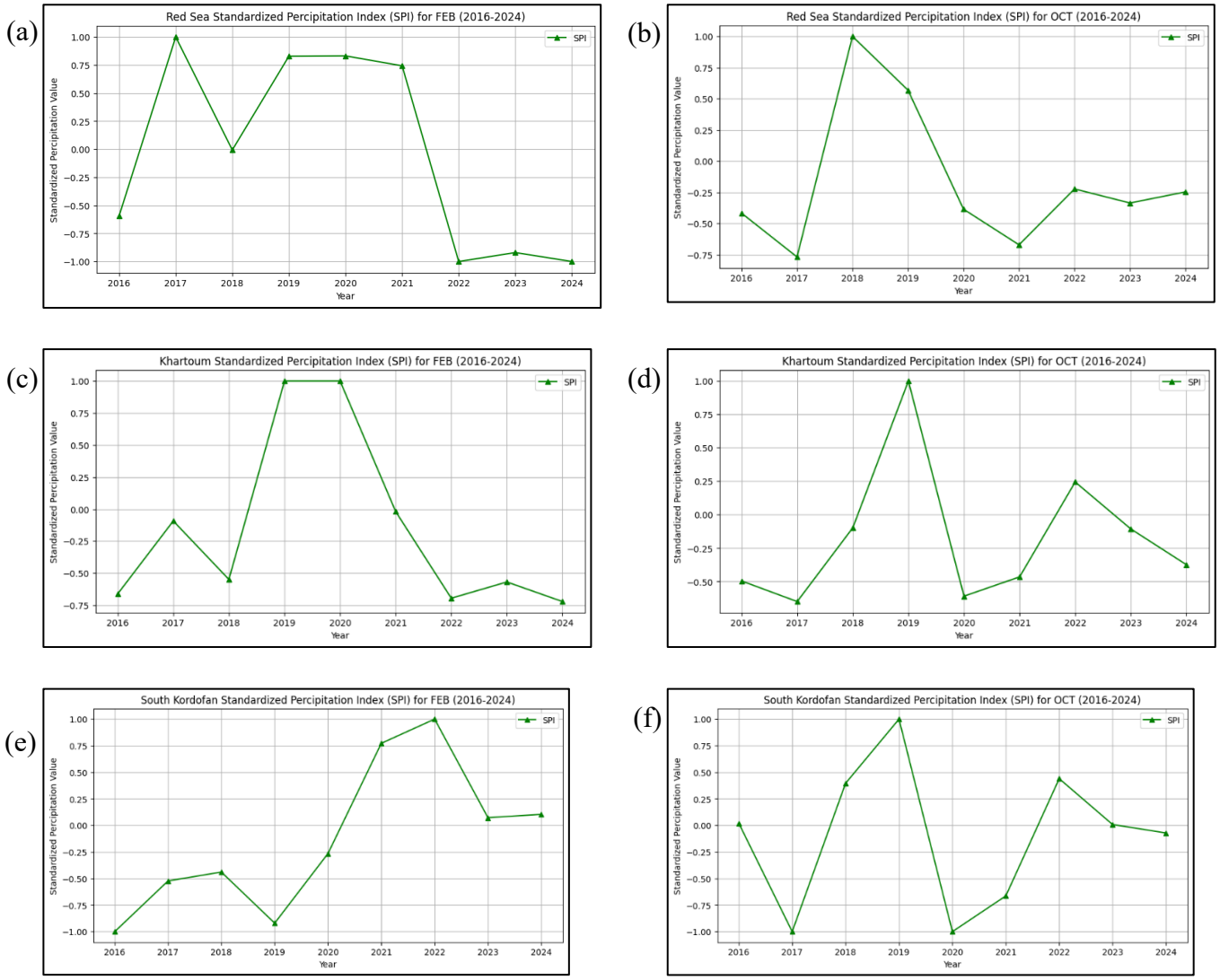

Figure 7

GPM Precipitation trends across 2016-2024 to validate MODIS and SMAP values. (a) Red Sea February SPI shows a wet peak in 2017, then shifts to persistent dryness after 2020. (b) Red Sea October SPI peaks in 2017, then remains near or below average from 2018 onward. (c) Khartoum February SPI rises to a 2018 peak before dropping sharply into dryness by 2022. (d) Khartoum October SPI is wet in 2017–2018, followed by alternating wet and dry years. (e) South Kordofan February SPI is strongly wet in 2017–2019, then turns dry by 2023–2024. (f) South Kordofan October SPI shows wetness in 2017, sharp decline in 2020, and low values after.

4.3 Seasonal Difference Analysis

Seasonal Difference analysis was conducted to examine if the relation between MODIS and SMAP is more accurate when both seasons are analysed rather than individually.

4.3.1 Across Satellite Missions

Figures 4, 5, and 6 are scatterplots extracted to evaluate how seasonal difference values of MODIS and SMAP alight. These graphs show R^2 values. R^2 score is very low for Red Sea and Khartoum indicating the regression model between MODIS and SMAP do not match eve with seasonal variation. In contrast, South Kordofan shows high relations between MODIS and SMAP seasonal difference values.

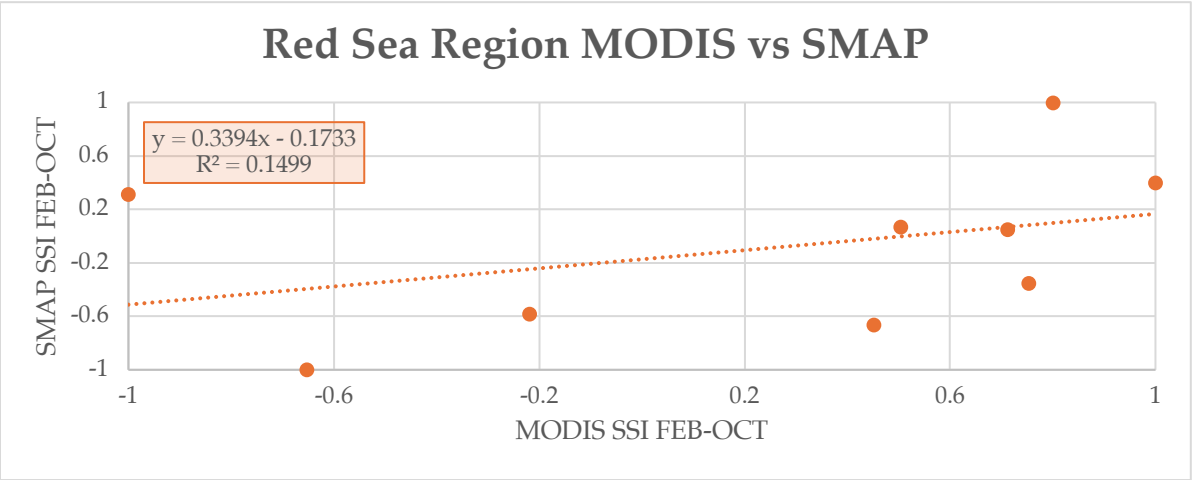

Figure 4

Seasonal difference SSI values between MODIS and SMAP for Red Sea region

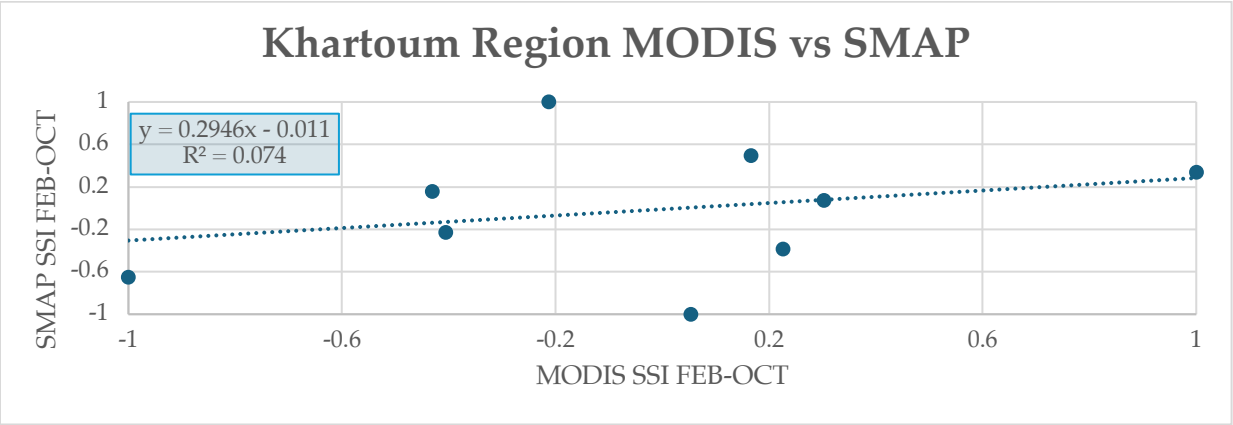

Figure 5

Seasonal difference SSI values between MODIS and SMAP for Khartoum regions

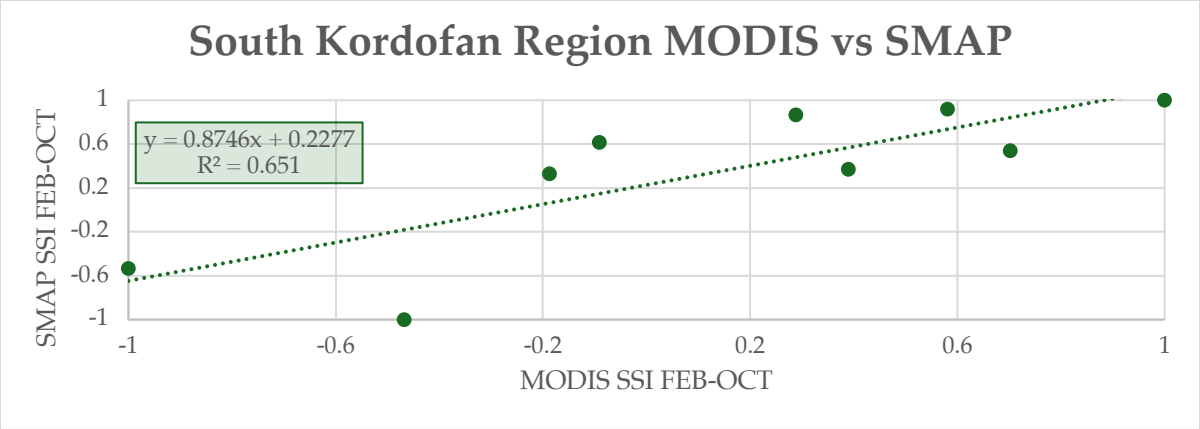

Figure 6

Seasonal difference SSI values between MODIS and SMAP for South Kordofan region

4.3.2 Across Regions

A cross-regional evaluations of seasonal variations in soil moisture from 2016 to 2024 is shown in Figure 7. Overall, soil moisture patterns throughout the study area are increasing when compared to 2016. This pattern is consistent with scope of the study area. Until around 2018, soil moisture values in Red Sea regions were comparatively low, yet over the same period, Khartoum showed a slight increase trend. For large portion of that time, south Kordofan continuously has more soil moisture than Khartoum and the Red Sea.

In contrast, by 2024, the trend shifts. The red sea show hight soil moisture than both Khartoum and South Kordofan. This is supported by the spatial patterns observed in 2024, as shown in section 4.1.1, further validating the regions differences in the times series graphs.

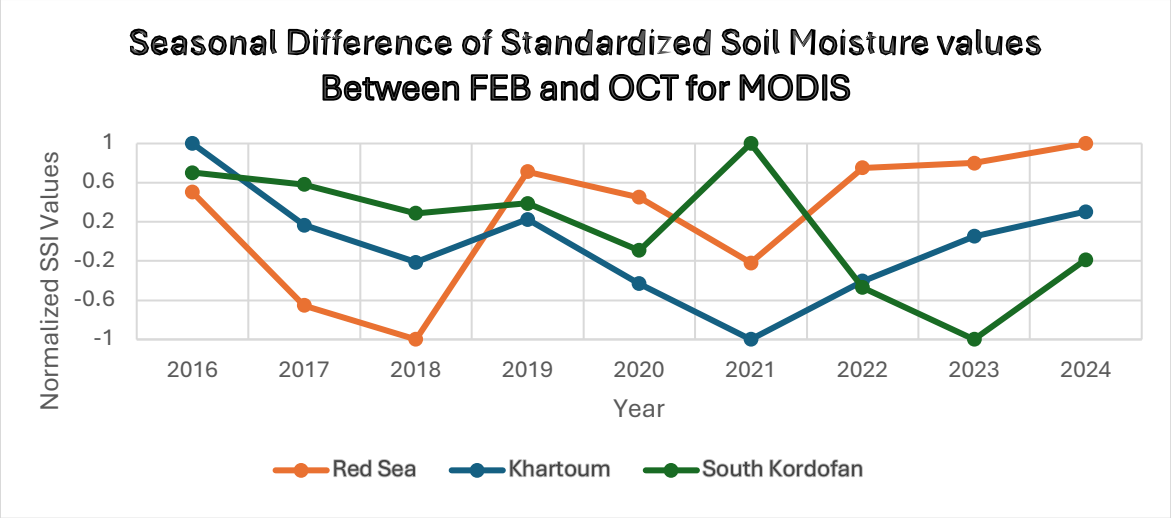

Figure 7

Shows seasonal difference for calculated SSI values from MODIS

Referring to Figure 8, the overall patterns show relatively increase in soil moisture content, but high variability compared to MODIS.

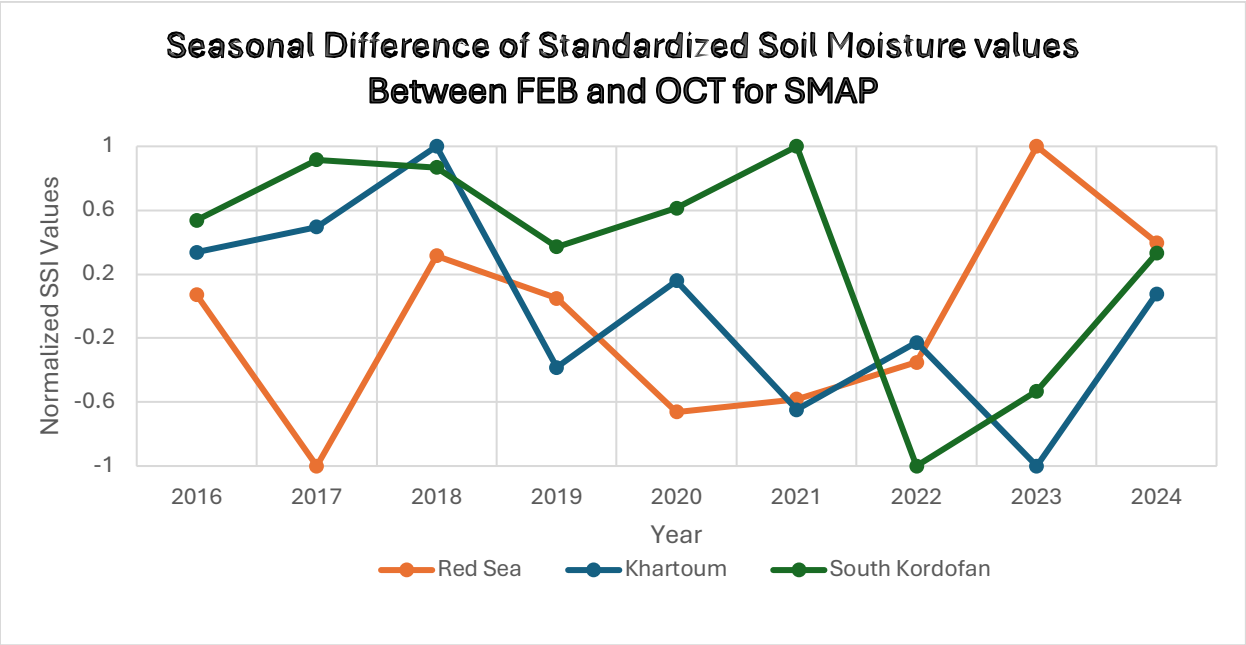

Figure 8

Shows seasonal difference for calculated SSI values from SMAP

4.3.3 Statistical Analysis

The statistical analysis compared to across satellite missions and across regions, with bias and correlation coefficient calculation were performed. Bias (Table 3) indicates how close MODIS values are to SMAP on an average. However, lower bias does not necessarily imply better agreement, as the relationship can still be weak for yearly changes captured. Correlation coefficient (Table 4) indicates overall strength of the MODSI values linear relation to SMAP values. Higher the correlation stronger the relationship between the values.

For Red Sea, the average bias shows moderate positive, meaning that MODIS generally overestimates SMAP values on an average. Similarly, the correlation coefficient demonstrates moderate relationship between MODIS and SMAP. Overall, there is moderate variations in soil moisture suggesting MODIS estimates seasonal variation with moderate accuracy.

In Khartoum, the bias is very low negative, indicating that on an average, MODIS values are similar to SMAP values. Whereas the correlation coefficient reveals weak relationship, indicating the yearly estimation of values from MODIS does not alight with SMAP. This is supported by R^2 in section 4.3.1. this implies there is very high variation in soil moisture and does not accurately detect changes.

Lastly, for South Kordofan, the bias is low negative, indicating that MODIS slightly underestimates SMAP values on an average. However, the strong correlation indicates that MODIS values alight strongly with SMAP pattern closely, at lower level which is supported by time series graphs. This indicates low variation in soil moisture and high accuracy in detecting changes.

Table 3

Bias analysis across regions for the normalized seasonal difference between MODIS and SMAP

Bias	Year	Red Sea	Khartoum	South Kordofan
	2016	0.433034859	0.66230834	0.163757238
	2017	0.346885514	-0.3295028	-0.335720307
	2018	-1.315568982	-1.2128041	-0.58020263
	2019	0.662807583	0.60853964	0.017713393
	2020	1.114040337	-0.5905326	-0.705844324
	2021	0.362900818	-0.3527445	0
	2022	1.104429498	-0.1796065	0.531579026
	2023	-0.198815476	1.05382672	-0.469072091
	2024	0.602276728	0.22676204	-0.519208075
Average		0.345776764	-0.0126393	-0.21077753

Table 4

Correlation coefficient for normalized seasonal difference values of MODIS and SMAP

Correlation	Year	Red Sea	Khartoum	South Kordofan
	2016-2024	0.387126501	0.27194392	0.806846059
		Moderate	Weak	Strong

5. Discussion

5.1 Results Interpretation

Overall, the results emphasise the variation between regional climate impact the values retrieved from satellites. Throughout the analysis SMAP has shown to be more accurate in the

month of October due to the wet season. SMAP is sensitive to dry season as the values between precipitation and soil moisture was inverse. MODIS does estimate to SMAP values in the vegetated regions such as South Kordofan, where the correlation was relatively high. The reason is as the vegetation increase the LST changes and MODIS will be able to predict accurately.

The relationship between soil moisture and precipitation is inverse rather positive most likely because of two processes: river transport and expansion brought on by upstream precipitation and hydraulic redistribution by tropical trees (Shelliot et al., 2018). Examining the graphs, the relationship is inverse since the Red Sea state borders a major river (Red Sea). However, because Khartoum has s smaller river (Blue Nile) and is less affected, it exhibits a relatively positive association between soil moisture and precipitation. Finally, there are no water bodies in South Kordofan or nearby and has moderately favourable relationship.

5.2 Previous Work

Despite the evolution since 1972, retrieving soil moisture has its own set of drawbacks, even with the development of remote sensing technology. Although they offer large-scale soil moisture data, SMAP has coarse resolution will restricts their use in diverse setting (Petropoulos et al., 2015). In-situ measurements to validates MODSI and SMAP data combined would be more accurate soil moisture estimation. Additionally, the use of machine learning models shows efficient at managing high-dimensional data (Gaona et al., 2022). However, there are further studies being done to accurately retrieve soil moisture from various application (Le & Liou, 2021).

5.3 Accuracy Errors

The accuracy of soil moisture retrieval is greatly impacted by vegetation, especially when passive microwave sensors and optical sensors are used. Dense vegetation can mask soil signals causing moisture levels to be overestimate or underestimated. For example, South Kordofan in the south part of Sudan, the MODIS SSI values were underestimated compared to SMAP values. Seasonal variation in soil surface impacts reflectance characteristics that sensors capture. To overcome these challenges calibration to large amount of ground truth data is necessary (Yang et al., 2022). Furthermore, due to the difference in porosity and organic content that can change the dielectric characteristics that microwave sensors detect, homogeneity in soil texture also impact soil moisture retrieval (Huber et al., 2011). For example, Red Sea region show high variability. Lastly, errors mainly could also be due to the spatial and temporal resolution difference between MODIS and SMAP. Upscaling MODIS refined the errors but may not accurately reflect SMAP soil moisture. SMAP is known to measure accurate soil moisture at root-zone level rather than surface, which drawback when combining datasets for soil moisture analysis.

6. Conclusion

This study demonstrates the utility of multi-sensor satellite data in assessing soil moisture variability and LULC change across semi-arid regions of Sudan. Both MODIS and SMAP

datasets successfully captured the interannual oscillations of soil moisture, with notable dry anomalies in 2017 and 2024 and wet anomalies in 2018 and 2021. Regional analyses revealed that Khartoum experienced pronounced seasonal oscillations, while South Kordofan showed persistent drying in recent years. These findings provide evidence of the increasing vulnerability of Sudan's landscapes to climate variability and land degradation.

The integration of indices such as SSI along with statistical validation, supports the reliability of remote sensing methods for long-term monitoring. However, challenges remain in addressing differences in anomaly magnitudes between datasets and ensuring consistent calibration across spatial scales. Future research should extend the temporal coverage, integrate ground-based measurements for validation, and explore socio-economic drivers of land change. Strengthening satellite-based monitoring will be essential for supporting sustainable resource management and guiding regional adaptation strategies.

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